1.

List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

ANS:-

1. IP Address Classes (A, B, C, D, E)

IP Class	First Octet Range	Purpose	Real-World Example
Class A	1–126*	Large networks	A large enterprise network using a private Class A range such as 10.0.0.0.
Class B	128–191	Medium-sized networks	A college network using the private range 172.16.0.0 to 172.31.255.255.
Class C	192–223	Small networks	A home router using 192.168.1.0/24.
Class D	224–239	Multicast communication	A video-streaming service sending data to a multicast group.
Class E	240–255	Reserved for experimental or future use	A networking laboratory conducting experimental IP tests.

Note: 127.x.x.x is reserved for loopback, so it is not used for ordinary Class A host addressing. Classes A, B, and C are the traditional unicast classes; modern networks generally use CIDR (Classless Inter-Domain Routing).

2.

Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Hint: Show your working steps for each calculation.

ANS:-

Calculate Network Address, Broadcast Address, and Valid Host Range

Given:

- IP address: 192.168.10.45
- Subnet mask: 255.255.255.0
- CIDR notation: /24

Step 1: Find the network address

The subnet mask /24 means the first 24 bits identify the network, while the last 8 bits identify the host.

IP address: 192.168.10.45

Subnet mask: 255.255.255.0

The last octet is the host portion. Set it to 0 to find the network address.

Network address = 192.168.10.0

Step 2: Find the broadcast address

Set all host bits to 1, making the last octet 255.

Broadcast address = 192.168.10.255

Step 3: Find the valid host range

- First valid host = Network address + 1
- Last valid host = Broadcast address − 1

Therefore:

- First valid host: 192.168.10.1
- Last valid host: 192.168.10.254

Final Answer

Network address	192.168.10.0
Broadcast address	192.168.10.255
Valid host range	192.168.10.1–254
Usable hosts	254

3.

Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

ANS:-

Subnetting Plan for a Food Delivery Startup

Given network: 10.0.0.0/24

We need four separate subnets for Delivery Partners, Restaurants, Customers, and Admins.

Step 1: Calculate the required subnet bits

To create four subnets:

$$2^n \geq 4$$

$$2^2 = 4$$

Therefore, borrow 2 host bits from the original /24 prefix.

New prefix = /24 + 2 = /26

New subnet mask = 255.255.255.192

Each subnet contains 64 total addresses, of which 62 are usable host addresses.

Step 2: Assign the subnets

Department	Network Address	Subnet Mask	Valid Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1–10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65–10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129–10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193–10.0.0.254	10.0.0.255

Conclusion: The /24 network is divided into four equal /26 subnets. Each subnet supports 62 usable IPv4 addresses, and the separation helps organize network access and security policies.

4.

You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Constraint: Show your calculation steps and final answers.

ANS:-

Subnetting a College Wi-Fi Network

Given:

- Original network: 172.16.0.0/16
- Required subnets: At least 20

Step 1: Calculate the number of borrowed bits

Use the formula:

$$2^n \geq 20$$

- 2^4 = 16, which is insufficient.
- 2^5 = 32, which is sufficient.

Therefore, borrow 5 bits from the host portion.

Step 2: Calculate the new subnet mask

New prefix=16+5= /21

A /21 subnet mask in binary is:

11111111.11111111.11111000.00000000

In decimal:

255.255.248.0

Step 3: Calculate the number of subnets

$$2^5 = 32 \text{ subnets}$$

The network can support 32 subnets, satisfying the requirement of at least 20.

Step 4: Calculate usable hosts per subnet

The number of remaining host bits is:

32-21=11

Total addresses per subnet:

2^11=2048

Subtract the network and broadcast addresses:

2048-2=2046

Final Answer

New subnet mask	255.255.248.0
CIDR prefix	/21
Total subnets	32
Usable hosts per subnet	2,046

5.

A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

ANS:-

Why Is 192.168.1.300 an Invalid IP Address?

An IPv4 address consists of four octets separated by dots. Each octet must have a decimal value between 0 and 255.

The given address is:

192.168.1.300

The last octet, 300, exceeds the maximum permitted value of 255. Therefore, this is an invalid IPv4 address.

For a typical /24 home network, valid host addresses include:

- 192.168.1.1

- 192.168.1.50
- 192.168.1.200
- 192.168.1.254

The usual host range is 192.168.1.1 to 192.168.1.254, assuming subnet mask 255.255.255.0.

Conclusion: Replace 300 with a valid octet from 0 to 255, ensuring that the resulting address is an available host address within the correct subnet**.**